

Harsh Shrivastava

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Summary

AI & Automation Developer with 4+ years of engineering experience, specializing in Copilot Studio agents, Power Platform, and LLM-based automation. Currently at Ernst & Young (EY), building Copilot Studio agents and Power Platform solutions across corporate functions for the Immigration Control System, alongside biometric integrations for Government of India projects.

Skills

AI & Automation: Copilot Studio, Power Automate, Power Apps, RAG, Prompt Engineering, Evaluation, Claude, GitHub Copilot, LangChain, MCP

Core Tech Stack: Python, React.js, Next.js, Node.js, FastAPI, Dataverse

Programming Languages: TypeScript, JavaScript, Python, Kotlin, Java, SQL, Dart

Tools & Architecture: Docker, Git, IAM, SSO, REST APIs, IPC Architecture, Firebase, Postman, Jira

Experience

Ernst & Young (EY)

Feb 2025 – Present

Associate Consultant - Technology — Delhi, India

- Work as Power Platform Developer for the Immigration Control System, building Copilot Studio agents across corporate functions (visa processing, biometric exception handling, case management) to automate query resolution and cut manual turnaround.
- Built a Copilot Studio agent for the Visa Portal enabling applicants to query application status and biometric appointment status via natural language, integrated with Dataverse and REST APIs through a controlled gateway layer.
- Designed a Power Apps model-driven application on Dataverse for internal corporate operations, covering case escalation tracking and multi-level approval workflows with Power Automate-driven notifications.
- Developed a custom AI agent (Azure AI Foundry) to accelerate internal module development, generating boilerplate scaffolding and surfacing relevant internal documentation on demand, cutting new-module ramp-up time for the team.
- Led a 5-member React.js team for the Immigration Control System core platform; designed clean UI-to-service boundaries with state synchronization and error handling.
- Build a Tauri desktop app (Rust + TypeScript) for secure SFTP file transfer in air-gapped environments, using Electron-equivalent Inter-Process Communication (IPC) architecture with message schemas and backpressure handling.
- Integrated biometric hardware (Page Tray Readers) with streaming output, retry logic, and error recovery, supporting 10,000+ passengers per day.
- Resolved 100+ production issues by adding logging and tracing observability, reducing turnaround time by 25%; optimized NIC pipelines, cutting processing time by 40%.
- Built a Face Match and Face Detection proof-of-concept using CNN-based face recognition, exposed via a FastAPI REST API service (Uvicorn) for real-time similarity scoring and identity verification.

Z1 Tech

Feb 2023 – Feb 2025

Software Developer — Gurugram, Haryana

- Published Flapster and Pixel Perfect end-to-end, scaling to 15,000+ users in 6 months.
- Built an Ads Mediation SDK (Kotlin) with clean UI/backend/third-party boundaries, boosting ad revenue by 20%.
- Developed React.js/Next.js dashboards with performance profiling, cutting load time by 30%.
- Optimized company websites for performance, growing organic traffic by 12%.

Demotic Technologies Pvt. Ltd.

Aug 2022 – Feb 2023

Mobile App Developer — Remote

- Delivered 5+ features and resolved 50+ production bugs in 3 months.

- Improved FCM notification reliability with retry logic, raising success rate by 35%.
- Added crash/error observability, cutting crash rate from 6% to below 2%.

Projects

Copilot Studio Agent — Visa Application Assistant — Copilot Studio, Power Platform, Dataverse

- Built a Copilot Studio conversational agent on the public-facing Visa Portal, enabling applicants to query application status, biometric appointment status, and case history via natural language.
- Integrated with Dataverse and REST APIs exposed through a controlled gateway layer, retrieving status data from internal ICS systems.

Copilot Studio Agent — Biometric Exception Triage — Copilot Studio, Power Automate

- Designed a Copilot Studio agent to triage Page Tray Reader biometric verification exceptions, classifying error types and routing cases to the correct resolution workflow.
- Connected the agent to Power Automate flows for automatic ticket creation and status updates, reducing manual triage effort for support staff.

Power Apps Case Management System (ICS) — Power Apps, Dataverse, Power Automate

- Built a Power Apps model-driven application on Dataverse for internal ICS corporate operations, covering officer shift assignments, escalation case tracking, and multi-level approval workflows.
- Implemented role-based access and automated status notifications via Power Automate, streamlining coordination across immigration control teams.

Custom RAG Pipeline — Azure OpenAI, Azure AI Search, Python

- Built a Retrieval-Augmented Generation (RAG) pipeline using Azure OpenAI and Azure AI Search, indexing internal documentation for natural-language Q&A with citation-backed responses.
- Implemented chunking, embedding generation, and hybrid (vector + keyword) search to improve retrieval relevance over naive semantic search.

Azure AI Foundry Agent — NIC Pipeline Observability Assistant — Azure AI Foundry, Azure OpenAI, Python

- Prototyped an Azure AI Foundry agent to summarize NIC pipeline logs and error traces, surfacing likely root causes and suggested remediation steps for recurring processing failures.
- Used function calling to connect the agent to internal logging/observability APIs, reducing manual log triage time during production issue investigation.

Education

Jaypee University of Engineering and Technology — B.Tech in Computer Science and Engineering (Artificial Intelligence) 2023

CGPA: 8.4

Achievements

Extraordinary Award, Ernst & Young — Recognized for exceptional performance and high-impact contributions to the NIC and Immigration Control System projects.